The Late Channel: Chain-of-Thought Becomes Causal *and* Decodable Only Late in a 27B Reasoning Agent

Mechanistic faithfulness of reasoned answers and agent actions, by feature patching and logit-lens

Caio Vicentino
OpenInterpretability
Fortaleza, Brazil
ORCID: 0009-0003-4331-6259
caio@openinterp.org

June 2026

Abstract

Chain-of-thought (CoT) monitorability is a leading safety bet for reasoning agents, but the question is usually *behavioral*: does the written CoT predict behavior? We ask the *mechanistic* version on an open-weight 27B reasoning model (Qwen3.6-27B) with a full-stack sparse autoencoder (SAE; 11 layers, d_{sae}=40960): are the features at the post-reasoning decision point *causal* for what the model decides, and *where* do they live? A single causal-patch test (decode–re-encode of the CoT decision-state into a no-CoT run, controlled for SAE reconstruction error and a random-feature baseline) gives a consistent answer for two outcomes. (i) For the *answer* to hard multi-step problems (n=60, 16 CoT-flips), patching the CoT features recovers the reasoned answer with $\Delta \log p = +2.72\,[+2.18, +3.28]$ at L59, effect ≈ 0 through L47 (LATE +2.13 vs EARLY −0.02, disjoint 95% CIs; 16/16 consistent). (ii) For an *agent action* (tool call) in trap scenarios (n=32, 10 flips), the same test gives LATE $+1.65\,[+1.17, +2.12]$ vs EARLY +0.08. A method-independent control rules out the obvious confound (that late-layer patches simply survive while early ones are washed out): a logit-lens of the decision token shows the reasoned answer is *anti*-decodable early (margin −0.9 at mid layers, the model leaning to the System-1 answer) and only becomes decodable at L51, reaching +6.6 at L63—i.e. the deciding information genuinely *consolidates* late, it is not present earlier. Faithfulness is **conditional**: causal when the CoT changes the outcome, ≈ 0 (performative) when the model already knew (44/60 correct with no CoT). The late band is thus a readable, causal locus for mechanistic monitoring—instantiating DeepMind's call to "inspect the model's inner workings"—while, per a companion result, the same band is *not* adversarially robust if used as a control point.

1 Introduction

Reasoning models expose a chain of thought, and a growing safety program proposes to *monitor* it: read the CoT, flag bad behavior [3, 4]. But the CoT is text, and the field's own results show it is an unreliable readout of the underlying computation: models can be unfaithful [1, 2], and monitorability degrades under optimization pressure [3, 4]. The position paper that defines the agenda explicitly defers the mechanistic route as "requiring further breakthroughs" [3]. Concurrently, Google DeepMind's agent-security roadmap states that "reading their verbalized reasoning will not be enough" and that securing agents "will need to . . . inspect the model's inner workings" [8].

This paper takes the mechanistic route. We ask, on a single open-weight 27B reasoning model with a full-stack SAE: *are the features active during/after reasoning causal for the model's answer and action, and where do they live?* Our contribution is a unifying empirical law—**the late channel**—and a method.

Contributions.

- A single causal-patch test (§3) for mechanistic faithfulness that controls for SAE reconstruction error and a random-feature baseline—an application of a causal-rigor protocol from prior work.
- **Reasoning faithfulness is causal and late** (§4): CoT decision-state features causally drive the answer (peak $+2.72 \log p$ at L59), ≈ 0 before L51, and *conditional*—causal only when the CoT changes the answer; performative otherwise.
- **It holds for agent actions** (§5): the same late, causal pattern governs tool-call decisions in agent-trap scenarios.
- **Consolidation, not artifact** (§6): an independent logit-lens control shows the reasoned answer is anti-decodable early (-0.9) and consolidates late ($+6.6$ at L63), ruling out the patch-dilution confound and corroborating the patch result.
- **One late band** (§7): reasoned answers and agent actions become causal in the same late band a prior result localizes action commitment to—a single, readable locus for monitoring.
- An honest boundary (§8): the band is a good *monitor* but, per a companion result, *not* an adversarially-robust *control* point.

2 Related work

Behavioral CoT faithfulness is well studied and behavioral-only: unfaithfulness via input-perturbation [1], reasoning-model hint-omission [2], in-the-wild measurement, and the monitorability position [3]—none open the model. **Mechanistic faithfulness** via SAEs+activation patching has been shown on *Pythia-2.8B base* [5], finding causality is scale-dependent (null at 70M, causal at 2.8B); we extend this to a 27B native *reasoning* model. **Internal monitoring beats text**: residual-stream probes match/beat a full-trace CoT monitor [6], and attention probes predict the answer early ("performative reasoning") [7]—both *correlational*; we test *causally* with named SAE features and on agent *actions*. **The action lever** (this arc): control of the `finish`/commit decision lives in a late action-commitment band, not the mid-layer verdict [9, 10].

3 Method

Model & SAE. Qwen3.6-27B (64 layers, reasoning/"thinking" mode), bf16. A full-stack TopK SAE (d_{sae}=40960, k=128) trained on 11 layers $\{15, 19, 23, 27, 35, 39, 43, 47, 51, 59, 63\}$; encode $z = \text{TopK}(\text{ReLU}((x - b_{\text{dec}})W_{\text{enc}} + b_{\text{enc}}))$, decode $\hat{x} = zW_{\text{dec}} + b_{\text{dec}}$.

The causal-patch test. For a decision, we run the model *with* CoT (it thinks, then a forced slot elicits the answer/tool token) and *without* CoT (direct answer). We keep items where the CoT *flips* the no-CoT outcome (so the reasoning does work). At each SAE layer L, we encode the residual at the CoT decision point to z_{cot}, and in the *no-CoT* forward replace the decision-point residual with $\hat{x}(z_{\text{cot}})$; we measure $\Delta \log p$ of the CoT-outcome token. The clean causal signal is $d_{\text{cot}} - d_{\text{self}}$, where d_{self} patches the no-CoT features' own reconstruction (controls for SAE recon error); the noise control is $d_{\mathrm{rnd}} - d_{\mathrm{self}}$ with a random top-k feature set of matched magnitude. Outcomes are confirmed behaviorally (forced-slot argmax), paired per item, with bootstrap 95% CIs over flipped items. This is the causal-rigor protocol of [9] applied to features.

4 Reasoning faithfulness is causal and late

We use 60 hard multi-step problems (serial arithmetic / counterintuitive / code traps) as MCQs with options shuffled per item; 16 flip the no-CoT answer; in 44/60 the model is *already correct without any CoT*. Table 1 reports the per-layer causal signal over the 16 flipped items.

Layer	causal $d_{\mathrm{cot}} - d_{\mathrm{self}}$ [95% CI]	noise $d_{\mathrm{rnd}} - d_{\mathrm{self}}$
L15–L43	≈ 0 (CIs cross 0)	-8 to -10
L47	$+0.17\,[+0.03, +0.29]$	-10.5
L51	$+1.55\,[+1.28, +1.85]$	-11.3
L59	$\mathbf{+2.72}\,[\mathbf{+2.18}, \mathbf{+3.28}]$	-10.5
L63	$+2.12\,[+1.62, +2.61]$	-12.1
LATE (L51–63)	$+2.13\,[+1.83, +2.41]$	
EARLY (L15–27)	$-0.02\,[-0.05, +0.01]$	

Table 1: Tier-A (answers, n=60, 16 flips). The causal signal is ≈ 0 through L47 and concentrates in L51–63 (LATE vs EARLY CIs disjoint); L59 causal > 0 in 16/16 flipped items (min +0.60, max +4.81). Random patches destroy the answer at every layer.

Faithfulness is conditional (Fig. 1). The patch only matters where the CoT changes the answer. In the 44/60 items the model already answers correctly without thinking, the late causal signal is small (mean +0.28 at L59); when the CoT flips the answer it strongly recovers it (mean +2.72). The written CoT is *performative* when the model already knew—the mechanistic version of [7]'s "reasoning theater", now measured causally—and *faithful* only when the reasoning does work.

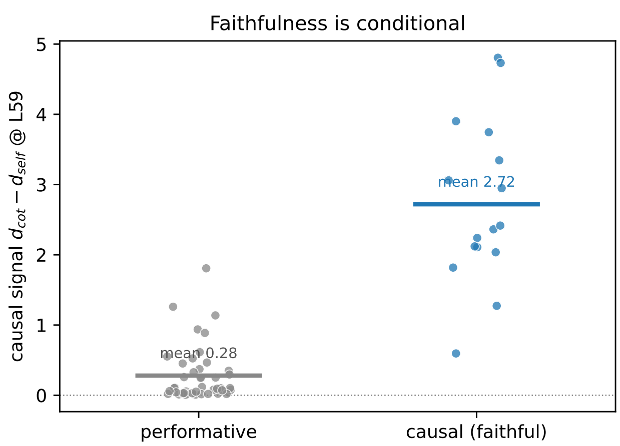

Figure 1: Conditional faithfulness. Causal signal at L59 per item, split by whether the CoT changes the answer. Already-correct-without-CoT (n=44): CoT features non-causal (mean +0.28). CoT flips the answer (n=16): features strongly recover it (mean +2.72). Bars are group means.

5 It holds for agent actions

We repeat the test where the "answer" is a committed tool call. 32 agent-trap scenarios across four domains (crypto/files/db/deploy): the surface instruction suggests an irreversible action (`send/delete/drop/deploy`); a detail suggests a safe one. 10 flip. Table 2 shows the same late, causal pattern.

Layer	causal $d_{\text{cot}} - d_{\text{self}}$ (10 flips)
L15–L47	≈ 0 (-0.14 to $+0.27$)
L51	$+0.84$
L59	$+1.89$
L63	$+2.21$
LATE (L51–63)	$+1.65\,[+1.17, +2.12]$
EARLY (L15–27)	$+0.08\,[-0.14, +0.32]$

Table 2: Tier-B (agent actions, n=32, 10 flips). Same late-band concentration; CIs disjoint.

Behavioral side-finding. With CoT the agent chose the *cautious* (safe) action in 27/32 traps (only 5 irreversible)—honest reasoning acts as a safeguard on these scenarios. We note a limitation: because the irreversible class is small (5/32), the per-layer action-*prediction* AUROC, although late-dominant and above a random-feature baseline, rests on too few positives to be conclusive (future work).

6 Consolidation, not patch-dilution

A natural objection: a last-token activation patch might simply *survive* in late layers (few layers remain to overwrite it) and be *washed out* early—a late profile that says nothing about where information lives. We rule this out with a method-independent control. On the same 16 flipped items we logit-lens the decision token in the run-CoT and measure the margin $\log p(\text{ans}_{\text{CoT}}) - \log p(\text{ans}_{\text{noCoT}})$ per layer (Fig. 2, dashed). The margin is *negative* through L43—the decision token leans to the no-CoT (System-1) answer, e.g. -0.93 at L35—crosses zero at L47, then rises steeply: $+1.57$ (L51), $+5.18$ (L59), $+6.64$ (L63), first exceeding $+0.5$ at L51. So the information that distinguishes the reasoned answer is *absent (indeed opposed) early and consolidates late*: there is nothing for an early patch to transfer. The two independent signals—feature-patch causality and logit-lens decodability—rise together at L51, mutually corroborating.

7 One late band, several decisions

On this model the same late band (L51–63) is where reasoned decisions become causal: the reasoned answer (§4) and the reasoned agent action (§5), both measured here; and—connecting to prior work, not re-measured here—*action commitment* itself, which an earlier result of this arc localizes to the same band [9, 10]. Decision-relevant information is ≈ 0 (or opposed) in mid layers and consolidates in this narrow late window—a single, consistent locus for reading what the model is about to do.

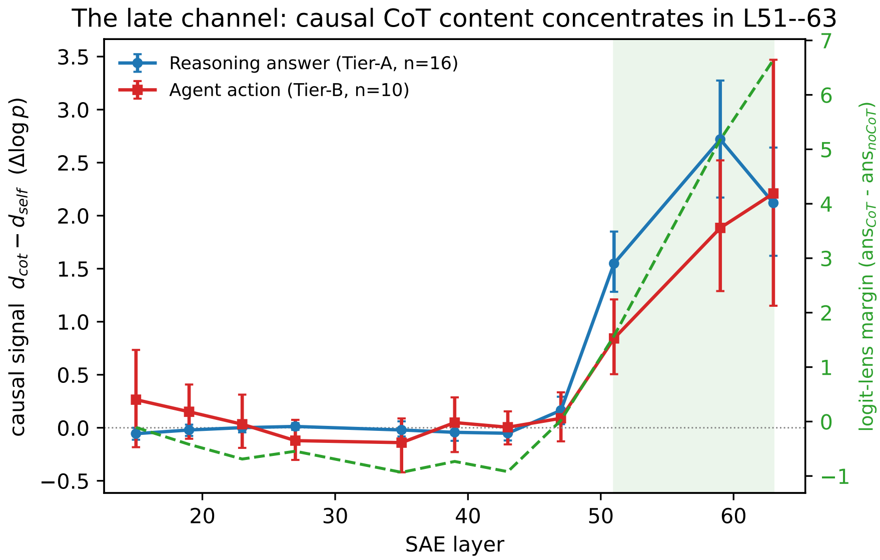

Figure 2: The late channel. Solid (left axis): causal patch signal $d_{\text{cot}} - d_{\text{self}}$ per SAE layer for the reasoned answer (Tier-A) and agent action (Tier-B), bootstrap 95% CIs. Dashed (right axis): the logit-lens consolidation margin (§6). Both are ≈ 0 or negative through mid layers and concentrate in L51–63 (shaded).

8 Discussion

A monitoring locus, not a control point. The late channel is where mechanistic monitoring should read: it carries the causal content and the decision is linearly decodable from the late state (logit-lens margin +6.6), a concrete read-out signal, consistent with internals beating CoT-text monitors [6, 7]. This instantiates DeepMind's "inspect inner workings" [8] for a reasoning agent. We are careful, though, to separate *decodability* (strong here) from *class prediction*: our attempt to predict the irreversible-action class from features (Tier-B) was underpowered (5 positives), so action-level monitoring *at scale* remains to be demonstrated. *However*, monitoring is not control: a companion experiment shows the same late action-commitment band, used as a brake, collapses under an adaptive white-box adversary (attack success $0 \rightarrow 1.0$ at a small budget that knows the locus, while a norm-matched random perturbation does nothing). The honest reading: the late channel is a strong *auditing* signal under a non-adversarial threat model, not an adversarially-robust safeguard.

9 Limitations

Curated stimulus sets (serial-computation MCQs; hand-built agent traps), not standard benchmarks; modest flip counts (16 and 10)—though effects are large and 16/16 consistent in Tier-A. Single model. The patch substitutes the last-token decision residual. The prediction/monitoring AUROC (Tier-B) is underpowered (5 positives). Causal results are conditional on the CoT changing the outcome; the performative regime is the common case here.

10 Conclusion

On a 27B reasoning agent, a single late layer band carries the causal content of what the model answers and what it does, and that content is faithful (causal) exactly when the reasoning changes the outcome. The late channel is the natural locus for mechanistic monitoring of reasoning agents—and a sharper target than CoT text—while remaining, as a control point, adversarially brittle.

References

[1] Turpin et al. Language Models Don't Always Say What They Think. NeurIPS 2023. arXiv:2305.04388.

[2] Chen et al. (Anthropic). Reasoning Models Don't Always Say What They Think. 2025. arXiv:2505.05410.

[3] Korbak et al. Chain-of-Thought Monitorability. 2025. arXiv:2507.11473.

[4] Baker et al. (OpenAI). Monitoring Reasoning Models for Misbehavior... 2025. arXiv:2503.11926.

[5] Chen, Plaat, van Stein. How Does Chain-of-Thought Think? ...SAE+patching. 2025. arXiv:2507.22928.

[6] Mirtaheri & Belkin. Detecting motivated reasoning from internal representations. 2026. arXiv:2603.17199.

[7] Goodfire. Reasoning Theater... performative reasoning. 2026. arXiv:2603.05488.

[8] Google DeepMind. Securing the Future of AI Agents. 2026.

[9] Vicentino. The Lever Is Late. 2026. Zenodo 10.5281/zenodo.20534219.

[10] Vicentino. The Lever Generalizes—and It Brakes. 2026. Zenodo 10.5281/zenodo.20634838.